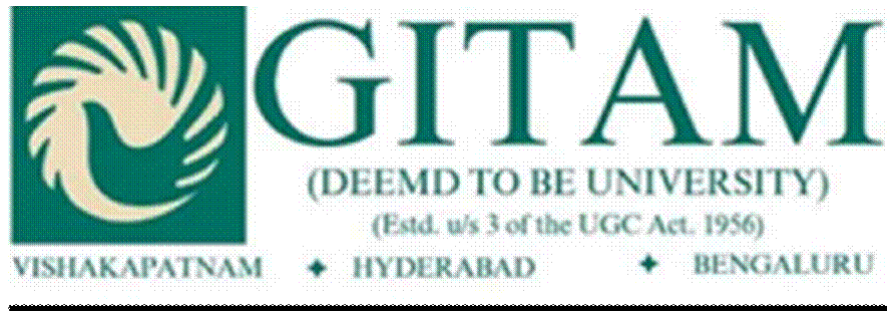




Department of Aerospace Engineering

FUNDING & SUPPORT PROPOSAL

IN-SPACE Model Rocketry India Student Competition 2026

2nd Edition | October–November 2026

Launch Site: Tumkuhiraj, Kushinagar, Uttar Pradesh

SUBMITTED TO

SUBMITTED BY

Date of Submission: [] | Academic Year 2025–2026

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Abstract

This proposal seeks institutional funding and sponsorship support from GITAM Deemed to be University under the Student Innovation Plan (SIP) for the participation of the Department of Aerospace Engineering's inaugural student team in the IN-SPACe Model Rocketry India Student Competition 2026 (2nd Edition), organized by the Indian National Space Promotion and Authorization Centre (IN-SPACe), Government of India.

This is a historic first in GITAM's academic journey — the Department of Aerospace Engineering is fielding its first competitive rocketry team at a nationally recognized, government-organized space technology event. The competition, conducted under the auspices of the Department of Space, Government of India, challenges undergraduate teams to design, fabricate, integrate, and launch a custom solid-propellant rocket capable of delivering a 1.0 kg satellite-class payload (CAN-7U) to a target altitude of 1,000 metres, with full telemetry, autonomous recovery, and post-flight analysis.

Participation in this competition directly aligns with GITAM's strategic vision of excellence in technology, innovation, and national contribution. It will provide students hands-on experience in aerospace systems engineering, propulsion, avionics, structures, and mission operations — skills that are critical to India's rapidly growing space sector.

The team respectfully requests a total funding support of ₹1,00,000 (Rupees One Lakh Only) from the institution, along with institutional facilitation in connecting with potential corporate sponsors. The funds will cover mandatory competition fees, workshop costs, component procurement, avionics development, structural fabrication, and ground station infrastructure.

1. About the Competition

1.1 Organizing Authority

The competition is organized by the Indian National Space Promotion and Authorization Centre (IN-SPACe) — a statutory body under the Department of Space, Government of India — in association with the Srinarayani Science Development Foundation (SSDF), Kushinagar, Uttar Pradesh. IN-SPACe serves as the single-window nodal agency for promoting and authorizing space activities by non-governmental entities in India.

Authority	Details
Parent Ministry	Department of Space, Government of India
Organizing Body	IN-SPACe — Indian National Space Promotion and Authorization Centre

Associate Body	Srinarayani Science Development Foundation (SSDF), Kushinagar, U.P.
Competition Edition	2nd Edition — IN-SPACe Model Rocketry India Student Competition 2026
Document Reference	IN-SPACe.MR.Model.CG.01, Issue No. 1, Dated 24 March 2026
Official Website	www.inspace.gov.in

1.2 Competition Overview & National Significance

IN-SPACe launched this competition to build India's future space workforce by challenging undergraduate students to develop real-world rocketry systems. The 1st Edition (2024–2025) witnessed approximately 70 teams from 18 states across India, with 33 teams reaching the Grand Finale at Kushinagar, Uttar Pradesh. Participating teams successfully deployed 1 kg payloads to altitudes up to 1 kilometre, demonstrating the competition's technical rigour and national reach.

The 2026 edition builds on this success with more structured phases, enhanced technical requirements, and a broader national participation base. The competition is designed to bridge the academia-industry gap in space technology and directly contributes to India's vision of becoming a global leader in the space sector under the National Space Policy 2023.

1.3 Primary Mission Statement

Each team must design and build a single-stage rocket capable of reaching a target altitude of 1,000 m (± 100 m) using a solid motor provided/procured from IN-SPACe-approved vendors, while safely carrying and recovering a 1.0 kg (± 0.05 kg) CAN-7U satellite-class payload with complete onboard telemetry, autonomous parachute recovery, and post-flight data analysis.

1.4 Core Technical Requirements at a Glance

Parameter	Specification
Target Altitude	1,000 m $\pm$ 100 m
Payload Mass	1.0 kg $\pm$ 0.05 kg (CAN-7U Satellite format)
Overall Rocket Length	Less than 180 cm (including fins)

Body Material Options	Aluminium (max 15 kg incl. motor) or PVC (~11.5 kg incl. motor)
Motor Diameter	94 mm (standard for both body types)
Maximum Total Impulse	2,800 Newton-seconds
Rocket Stages	Single stage only
Launch Angle	80° to 85° from horizontal
Descent Rate	2 to 5 m/s (± 0.5 m/s) — all parts tethered
Body Colour	Fluorescent (pink, red, or orange)
Structural Load (Launch)	Must survive 15g acceleration
Structural Load (Shock)	Must survive 30g shock
Motor Retention	Positive mechanical means (not friction fit)
Battery Type Allowed	Alkaline, Ni-Cad, Ni-MH, Lithium-ion (18650 type)
Prohibited Items	LiPo batteries, pyrotechnic separation devices, sharp edges
Team Size	5 to 8 undergraduate members
Teams per College	One (affiliated colleges may register separately)

1.5 Competition Phases

The competition is structured across seven progressive phases, ensuring systematic development and evaluation of each team's rocketry system:

Phase	Name	Description	Timeline
1	Registration	Team registration, fee payment, institutional approval	By 30 April 2026
2	Preliminary Design Review (PDR)	Design document submission and 10-minute jury presentation	1 May – 15 July 2026
3	Rocket Modelling Workshop	Mandatory 1-week IN-SPACe hands-on training; scaled rocket launch	1–16 August 2026
4	Critical Design Review (CDR)	Final design, component procurement begins; jury evaluation	17 Aug – 10 Oct 2026
5	Flight Readiness Review (FRR)	GO/NO-GO evaluation by jury; launch clearance	October–November 2026
6	Launch	Single launch opportunity; Grand Finale at Kushinagar, U.P.	October–November 2026
7	Post Flight Analysis (PFA)	Jury reviews flight data; winners announced (3–5 teams)	After competition

2. GITAM Team — Aerospace Rocketry Initiative (GARI)

2.1 Historical Significance

This participation marks a watershed moment for GITAM Deemed to be University. The Department of Aerospace Engineering is entering the IN-SPACe Model Rocketry Competition for the first time in the history of the institution. GITAM's GARI team will represent the university on a national government-organized stage, putting GITAM alongside premier engineering institutions from across India. This initiative is born under the Student Innovation Plan (SIP) and is entirely student-led.

2.2 Team Composition

S.No	Name	Institute ID	Year	Branch	Role in Team
1	Shaik adil ifthakar	2023003180	3rd	Aerospace	Team Leader , Analysis engineer
2	Harshith J V N P	2023001296	3rd	Aerospace	Co Lead, supportive engineer
3	Nischal Aditya M	202300	3rd	EECE	Communication engineer
4	Shaik farhat raheem	2025182947	1st	Aerospace	Design engineer
5	T Thrinadh	2023001876	3rd	Aerospace	Propulsion engineer
6	Achanta sri karthikeya	2025029155	1st	EECE	Electronics engineer
7	Mokshagna Mopidevi	2023002980	3rd	Aerospace	Computer engineer
8	Shreemann Narayana Karnakoti	2024036865	2nd	Aerospace	Structural engineer
9	[Faculty Advisor Name]	[Faculty ID]	—	Aerospace Engg.	Faculty Advisor (Point of Contact)

Note: The faculty advisor acts solely as the institutional point of contact and does not participate in design decisions, per IN-SPACE competition rules.

2.3 Alignment with Institutional Goals

Participation in IN-SPACE Model Rocketry 2026 directly advances GITAM's stated institutional goals of research excellence, student innovation, and societal impact. It aligns with:

- (a) GITAM's commitment to providing transformative undergraduate education with real-world engineering exposure.

- (b) The National Education Policy (NEP) 2020's emphasis on experiential, multidisciplinary, and project-based learning.
- (c) India's National Space Policy 2023, which encourages private and academic participation in space activities.
- (d) GITAM's aspiration to rank among India's leading aerospace engineering departments.

3. Proposal for Funding and Institutional Support

3.1 Proposal Overview

The GITAM Aerospace Rocketry Initiative (GARI) team respectfully submits this proposal to the following authorities for financial support and institutional facilitation:

Proposal Submitted To	Details
Primary Funding Authority	Dean of Research & Innovation, GITAM — under Student Innovation Plan (SIP)
Department Support	Head of Department, Aerospace Engineering, GITAM
Sponsorship Facilitation	GITAM Industry & Alumni Relations Office — to connect team with potential sponsors
Regulatory / Governing Body	IN-SPACe, Department of Space, Government of India
Competition Coordinator	Srinarayani Science Development Foundation (SSDF), Kushinagar, U.P.

3.2 Nature of Support Requested

3.2.1 Financial Support from GITAM (Student Innovation Plan)

The team requests a total grant of ₹1,00,000 (Rupees One Lakh Only) from the institution under the Student Innovation Plan. The complete detailed cost breakdown is provided in Section 4 (Bill of Materials & Budget). This grant will cover mandatory competition fees, workshop participation costs, structural and avionics components, and ground station development.

3.2.2 Institutional Facilitation for Sponsorship

In addition to direct financial support, the team requests that GITAM's Industry Relations and Alumni Affairs offices formally facilitate connections with potential corporate sponsors in the aerospace, defence, electronics, and manufacturing sectors. Specifically, the team requests:

- (a) An official GITAM endorsement letter to be shared with potential sponsors, confirming institutional support.
- (b) Introductions to companies under the GITAM Industry Partnership network — particularly firms in the aerospace, electronics, and precision manufacturing domains.
- (c) Permission to display sponsor logos on the rocket body, team uniforms, and all competition documentation, subject to IN-SPACE and institutional guidelines.
- (d) Assistance in drafting formal sponsorship agreements and acknowledgement protocols.

3.2.3 Departmental Support

The team requests the following non-financial support from the Department of Aerospace Engineering:

- (a) Access to the department's laboratory facilities for component testing, integration, and pre-launch simulation.
- (b) Faculty mentorship (up to 2 external mentors permitted per IN-SPACE rules, distinct from the faculty advisor).
- (c) Access to institutional computing resources for Finite Element Analysis (FEA) and flight simulation software.
- (d) Official institutional approval letter on GITAM letterhead, as mandated by IN-SPACE for registration.
- (e) Permission to dedicate a designated workspace for rocket assembly and integration activities.

3.3 Justification for Funding

Justification	Elaboration
Historic First	No team from GITAM has ever participated in a government-level national rocketry competition. This sets a precedent for GITAM's presence in India's space ecosystem.
National Visibility	GITAM's name will appear alongside IITs, NITs, and top engineering colleges at a government-organized national competition, significantly enhancing institutional prestige.
Alignment with IN-SPACE / Dept. of Space	Participation is in direct response to a Government of India initiative aimed at building India's future aerospace workforce — a national priority.
Student Skill Development	Students will gain hands-on experience in propulsion, avionics, structures, telemetry, and mission operations — directly applicable to ISRO, DRDO, and private space sector careers.

Research & Publications	The project will generate technical reports, design data, and potentially peer-reviewed publications — contributing to GITAM's research output.
NEP 2020 Compliance	The project exemplifies project-based, experiential, and multidisciplinary learning — fully aligned with the National Education Policy 2020.
Make in India	IN-SPACE explicitly rewards use of indigenous components. GITAM's participation promotes domestic manufacturing and innovation.
Low Investment, High Return	The total ask of ₹1,00,000 is modest for the level of national exposure, skill development, and institutional branding that participation will generate.

4. Bill of Materials & Detailed Budget

4.1 Mandatory Competition Fees, Motor & Propulsion Hardware

All government competition fees are GST-exempt. Motor procurement is mandatory from IN-SPACE approved vendors only; HAZMAT shipping costs are included in the ASP estimate.

S.No	Item	Payable To	Unit Cost (₹)	GST	Total w/ GST (₹)	Remarks
1	Competition Registration Fee (Phase 1)	SSDF — HDFC Bank, Gorakhpur	10,000	0%	10,000	Non-refundable; due by 30 Apr 2026
2	Rocket Modelling Workshop — Participant 1 (Phase 3)	IN-SPACE / Bharatkosh	25,000	0%	25,000	Mandatory 1-week training; government fee
3	Rocket Modelling Workshop — Participant 2 (Phase 3)	IN-SPACE / Bharatkosh	25,000	0%	25,000	Max 2 participants allowed per team
4	CDR Submission Fee (Phase 4)	SSDF	15,000	0%	15,000	Due at CDR submission; non-refundable

5	K-Class Motor Reload (IN-SPACe Approved Vendor)	Approved Vendor	19,000	18%	22,420	Single-use reload; 2800 N-s limit; HAZMAT shipping included
6	54mm Motor Casing Hardware (reusable)	Approved Vendor	20,750	18%	24,485	Includes casing, rear closure, retention tool
SUBTOTAL — Mandatory Fees & Propulsion					₹1,21,905	

4.2 Structural & Mechanical Components

Aluminium 6061-T6 path selected — preferred by IN-SPACe 'Make in India' evaluation criteria. Prices based on IndiaMART wholesale + 18% GST.

S.No	Component	Spec / Source	Qty	Unit Cost (₹)	GST	Total w/ GST (₹)	Remarks
1	Airframe Body Tube (94mm OD, 3mm wall)	Al 6061-T6 Pipe	6 kg	400/kg	18%	2,832	~2m length; precision finished
2	Fin Sheet Stock (3mm plate)	Al 6061-T6 Sheet	3 kg	480/kg	18%	1,699	For 4× trapezoidal/delta fins; machined to profile
3	Nose Cone (Ogive/Von Karman profile)	PETG/Nylon 3D Print	600 g	20/g	18%	14,160	3D printed at service bureau; 30g shock rated
4	Motor Mount Assembly + Centering Rings (94mm)	Al 6061-T6	1 set	2,500	18%	2,950	Positive mechanical retention; M12 bolt interface
5	Thrust Plate (Aluminium)	Al 6061-T6	1 pc	1,500	18%	1,770	M5 screw interface — 6 screws; M12 central bolt
6	Payload Bay Tube & Bulkheads	Al / G10 Fiberglass	1 set	3,000	18%	3,540	CAN-7U bay: 22.86 cm × 22.86 cm × ~22 cm
7	Rail Buttons / Launch Lugs (1010/1515 compatible)	Nylon/Al	2 pcs	350	18%	826	Match drilled; compatible with team launch rail

8	Shock Cord & Tether (Kevlar 400 lb test, 30 ft)	Braided Tubular Kevlar	1 set	2,500	18%	2,950	Heat-rated; all parts tethered; 30 ft length
9	Fluorescent Paint (1L enamel, orange/red/pink)	High-Gloss Enamel	2 L	450/L	18%	1,062	Mandatory fluorescent body colour — rules compliance
10	M5 Hex Head Bolt Box (Pack of 50)	SS Grade 8	1 box	471	18%	556	Body-to-thrust-plate interface fasteners
11	M12 SS Metric Bolts	SS Grade 8	10 pcs	10/pc	18%	118	Thrust plate central bolt; match drilled
12	Adhesives, Epoxies & Sealing Compounds	Araldite/JB Weld	1 set	1,200	18%	1,416	Structural fin bonding; bulkhead sealing
SUBTOTAL — Structural & Mechanical						₹33,879	

4.3 Payload System — CAN-7U Satellite (1 kg)

Stratologger CF is a mandatory GO/NO-GO item per competition rules. LiPo batteries strictly prohibited — only 18650 format with metal casing permitted.

S.No	Component	Spec / Source	Qty	Unit Cost (₹)	GST	Total w/ GST (₹)	Remarks
1	CAN-7U Satellite Chassis / Frame	Al Machined / Procured	1	5,000	18%	5,900	Must weigh 1 kg $\pm$ 0.05 kg; procured or fabricated
2	Payload COTS Backup Altimeter (Stratologger CF)	PerfectFlite COTS	1	7,700	18%	9,086	MANDATORY GO/NO-GO item; 9V; 24-bit sigma-delta
3	Payload Accelerometer (ICM-42688 / ADXL375)	3-Axis MEMS	1	1,200	18%	1,416	15g launch, 30g shock rated
4	Payload Telemetry — XBee Pro S2C (separate channel)	Digi XBee Pro S2C	1	3,300	18%	3,894	Separate NETID from main rocket; 1 km range

5	Payload Parachute & Deployment Mechanism (24")	Ripstop Nylon	1	2,350	18%	2,773	Descent rate 2–5 m/s for payload; 1.2 kg load
6	Payload Battery Pack — Li-ion 18650 (metal casing)	LG / Molicel 3500mAh	2 cells	750/cell	18%	1,770	LiPo STRICTLY prohibited; 30 min pad + flight
7	Environmental Sensor Module (Temp/Humidity/PM2.5)	I2C MEMS Sensor	1	1,500	18%	1,770	Optional experiment module; extra credit scoring
SUBTOTAL — Payload System (CAN-7U)						₹26,609	

4.4 Avionics & Electrical Systems

NavIC GNSS integration is strongly encouraged by IN-SPACe and carries evaluation scoring weight. Redundant barometric altimeters are mandatory. All wiring must use aviation-grade connectors with no spring contacts.

S.No	Component	Spec / Source	Qty	Unit Cost (₹)	GST	Total w/ GST (₹)	Remarks
1	Flight Computer (Teensy 4.1 / STM32)	ARM Cortex-M	1	3,000	18%	3,540	Flight state machine; data logging; recovery trigger
2	NavIC GNSS Receiver Module (L5/L1 dual-band)	SkyTraq S1216F8-G12 / Robu GNSS	1	4,500	18%	5,310	NavIC + GPS; encouraged by IN-SPACe; 1 km tracking
3	XBee Pro S2C Telemetry Module — Main Rocket	Digi XBee Pro S2C 63mW	1	3,300	18%	3,894	3.2 km LoS range; NETID = team number; 1 Hz min
4	Barometric Altimeter — Primary (BMP388 / MS5611)	MEMS Barometric	1	316	18%	373	High-res primary altitude sensor; I2C/SPI
5	Barometric Altimeter — Backup (BMP280)	MEMS Barometric	1	316	18%	373	MANDATORY backup; redundant recovery chain
6	3-Axis Accelerometer	High-g MEMS	1	1,200	18%	1,416	Rated 15g launch / 30g shock; flight data logging

	(ICM-42688 / ADXL375)						
7	SD Card Module + 32 GB MicroSD Card	SPI Module	1	500	18%	590	Mandatory onboard flight data storage per rules
8	Real-Time Clock Module (DS3231)	I2C RTC	1	350	18%	413	Mission timestamp; log synchronisation
9	Main Rocket Battery — 18650 Pack (2S or 3S)	LG / Molicel 3500mAh × 2	2 cells	750/cell	18%	1,770	LiPo prohibited; metal-cased 18650 only
10	Power Distribution PCB / BEC (5V/3.3V regulated)	Custom PCB / COTS BEC	1	1,500	18%	1,770	Regulated power to all avionics subsystems
11	Remote Ignition System (E-match compatible)	E-match + relay circuit	1	2,500	18%	2,950	External power switch + indicator light + GO/NO-GO
12	Avionics Bay Enclosure (hard-mounted)	Al / G10	1	1,000	18%	1,180	Shock-rated; shielded; hard-mounted via standoffs
13	Wiring Harness, Aviation-Grade Connectors, Standoffs	Misc. Electronics	1 set	1,200	18%	1,416	No spring contacts; shock-rated; redundant looms
14	Indicator LEDs, Buzzer & Arming Switch	Misc.	1 set	400	18%	472	Visual + audio ignition readiness; arming safety
15	Custom Avionics PCB (Power + Signal routing)	2-layer PCB fabrication	1	2,000	18%	2,360	Locally manufactured; reduces wire clutter
SUBTOTAL — Avionics & Electrical Systems						₹27,827	

4.5 Recovery System

Descent rate must be maintained at 2–5 m/s (± 0.5 m/s) for the full system. A 60" parachute is the minimum for a 15 kg aluminium rocket. All parts must be tethered together — no detached sections permitted.

S.No	Component	Spec / Source	Qty	Unit Cost (₹)	GST	Total w/ GST (₹)	Remarks
1	Main Parachute — 60" Ripstop Nylon (primary rocket)	Iris Ultra 60" / equivalent	1	17,250	18%	20,355	Descent 2–5 m/s for 15 kg full rocket; spill-hole design
2	Payload Recovery Parachute — 24" (CAN-7U)	Ripstop Nylon	1	2,350	18%	2,773	Separate payload recovery; 1 kg payload descent
3	Parachute Deployment Mechanism (CO2 / Spring ejection)	CO2 canister / spring	1	2,000	18%	2,360	No pyrotechnics; green propulsion only per rules
4	Kevlar Shock Cord — 400 lb test, 30 ft (main)	Braided Kevlar	1	2,500	18%	2,950	Heat-rated; tethers all parts to main parachute
5	Tether Line — Payload to Main Structure	Tubular Nylon / Dyneema	1	700	18%	826	Mandatory: all parts tethered together per rules
6	HAM Radio Beacon / GPS Tracker (post-landing locate)	APRS / LoRa tracker	1	3,000	18%	3,540	Optional — extra credit; aids recovery at 1 km distance
SUBTOTAL — Recovery System						₹32,804	

4.6 Ground Support Equipment (GSE) & Launch Infrastructure

The launch pad must be team-designed and fabricated. Launch rail, blast deflector, and support legs are mandatory for a K-class motor ignition. All GSE costs include 18% GST.

S.No	Component	Spec / Source	Qty	Unit Cost (₹)	GST	Total w/ GST (₹)	Remarks
1	XBee Pro S2C Ground Station Radio Unit	Digi XBee Pro S2C 63mW	1	3,300	18%	3,894	Paired with rocket XBee; NETID matched to rocket
2	Directional / Yagi Antenna (hand-held, elevated)	2.4 GHz Yagi	1	1,500	18%	1,770	Elevated for 1 km+ gallery coverage; gain ~10 dBi

3	USB Hub + Interface Cables	USB 3.0 Hub	1 set	400	18%	472	Data transfer + jury presentation post-flight
4	Ground Station Software Development (custom UI)	Python / Node.js	1	2,000	0%	2,000	Real-time altitude/velocity plot; CSV export; SI units
5	Portable Power Bank (min. 2 hr ground station ops)	20,000 mAh, 18650-based	1	2,000	18%	2,360	Standardised 18650 cells — shared with rocket charger
6	4040 T-Slot Aluminium Extrusion — Launch Rail (6 m)	MISUMI / IndiaMART 4040	6 m	650/m	18%	4,602	Team-fabricated launch pad; K-class motor rated
7	Blast Deflector Plate (steel, K-class rated)	Steel plate / Estes equiv.	1	1,300	18%	1,534	Mandatory for K-class ignition thermal load
8	Launch Rail Support Legs & Base Frame Hardware	Steel / Al angle iron	1 set	2,000	18%	2,360	80°–85° adjustable launch angle; rule compliant
9	Fire Extinguisher (CO2 type, 2 kg)	Safety Equipment	1	1,200	18%	1,416	Mandatory safety equipment at launch site
SUBTOTAL — Ground Support Equipment						₹20,408	

4.7 Fabrication & Labor (3D Printing + Electronics Assembly)

3D-printed nose cone uses PETG/Nylon filament rated to survive 30g shock. Electronics assembly labor covers avionics bay hard-mounting and wiring loom fabrication.

S.No	Item	Rate / Spec	Duration / Qty	Unit Rate (₹)	GST	Total w/ GST (₹)	Scope
1	3D Printing — Nose Cone & Aerodynamic Fairings	PETG/Nylon — ₹20/g	600 g	20/g	18%	14,160	Ogive/Von Karman nose cone; fin fairings; 30g shock rated
2	Electronics Assembly & Soldering Labor	Technician — ₹150/hr	20 hr	150/hr	18%	3,540	Avionics bay assembly; hard-mount standoffs; wiring loom

SUBTOTAL — Fabrication & Labor	₹17,700	
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4.8 Testing, Prototyping & Contingency

Phase 3 requires a scaled prototype rocket launch (~100 m). Vibration and shock qualification tests (15g/30g) are mandatory for CDR approval. Contingency covers motor HAZMAT shipping variance and vendor price fluctuations.

S.No	Component	Spec / Source	Qty	Unit Cost (₹)	GST	Total w/ GST (₹)	Remarks
1	Scaled Prototype Rocket (Workshop requirement — Phase 3)	Mixed materials	1	3,000	18%	3,540	Must reach ~100 m; payload nosecone tethered per rules
2	Parachute Ejection Testing Materials	CO2 / spring test rigs	1 set	1,500	18%	1,770	Deployment functional qualification tests
3	Vibration & Shock Test Materials (15g / 30g qual.)	Shaker table / materials	1 set	2,000	18%	2,360	CDR Phase 4 qualification test per competition rules
4	Contingency Reserve (~10% of total hardware)	—	—	—	—	~28,000	Unforeseen procurement; vendor price variance; re-tests
SUBTOTAL — Testing, Prototyping & Contingency						₹35,670	

4.9 Consolidated Budget Summary — True Engineer's Estimate

The following table consolidates all eight budget categories. The grand total of ₹3,16,802 clears the ASP market floor of ₹3,11,084 and represents the minimum credible all-in cost for competition-compliant participation. The SIP grant request of ₹1,50,000 covers mandatory fees and propulsion hardware in full, with the remaining ₹1,66,802 to be sourced via corporate sponsorship.

S.No	Budget Category	Subtotal (₹)	Notes
1	Mandatory Fees + Propulsion Hardware & Motor	₹1,21,905	Govt fees (GST-exempt) + K-class motor system
2	Structural & Mechanical Components	₹33,879	Al 6061-T6 airframe, fins, hardware, adhesives

3	Payload System — CAN-7U Satellite	₹26,609	Stratologger CF (mandatory GO/NO-GO item) included
4	Avionics & Electrical Systems	₹27,827	NavIC GNSS, XBee, sensors, custom PCBs
5	Recovery System	₹32,804	60" main chute, Kevlar cord, deployment mechanism
6	Ground Support Equipment	₹20,408	Launch rail, GSE, antenna, power bank, software
7	Fabrication & Labor (3D Print + Electronics Assembly)	₹17,700	3D-printed nose cone + avionics assembly labor
8	Testing, Prototyping & Contingency (~10%)	₹35,670	Prototype rocket + qual. tests + contingency buffer
GRAND TOTAL — Engineer's Estimate (incl. GST)		₹3,16,802	
Amount Requested from GITAM — Student Innovation Plan (SIP)		₹2,00,000	
Amount to be Raised via Corporate Sponsorships (GITAM facilitation)		₹1,16,802	

* All prices include GST at the applicable rate (0% on government fees; 18% on electronics, materials, and services). Prices are based on market data as of April 2026.

* The ASP tier of ₹3,11,084 from the engineering analysis (200+ vendor sources) is the absolute minimum compliant budget. This document's total of ₹3,16,802 reflects the realistic cost after confirming all mandatory items.

5. Project Timeline & Milestones

Phase	Activity	Dates	Deliverable	Evaluation
1	Registration & Institutional Approval	By 30 Apr 2026	Registration + ₹10,000	Team ID assigned

2	PDR Submission & Presentation	1 May – 15 Jul 2026	PDR Document	Jury graded
3	Rocket Modelling Workshop (IN-SPACe)	1–16 Aug 2026	Scaled rocket launch	Quiz + hands-on
4	CDR Submission & Review	17 Aug – 10 Oct 2026	CDR Document + ₹15,000	Jury graded
5	Flight Readiness Review (FRR)	Oct–Nov 2026	GO/NO-GO clearance	Jury inspection
6	Grand Finale Launch	Oct–Nov 2026	Single launch	Flight performance
7	Post Flight Analysis & Awards	Post-launch	Telemetry CSV + analysis	Final ranking

6. Requirements Compliance Matrix

The following matrix summarizes GITAM GARI team's compliance with all mandatory IN-SPACe competition requirements. Any deviation will be clearly documented in the PDR/CDR with technical justification.

S.No	Requirement	Compliance	Notes
1	Target altitude: 1,000 m $\pm$ 100 m	Compliant	Flight simulations will validate; OpenRocket/RASAero
2	Payload mass: 1.0 kg $\pm$ 0.05 kg (CAN-7U)	Compliant	Payload mass verified during integration
3	Single stage rocket only	Compliant	Design is single stage
4	Overall length < 180 cm incl. fins	Compliant	Target ~150 cm; within limit

5	Motor from IN-SPACe approved vendor only	Compliant	Will procure from listed vendors
6	Total impulse $\leq 2,800$ Ns	Compliant	Vendor motor specs to be verified
7	Body colour: fluorescent pink/red/orange	Compliant	Fluorescent orange selected
8	Team ID in English, Hindi & regional language	Compliant	Will include Telugu (regional)
9	No pyrotechnic separation devices	Compliant	Mechanical/CO2 deployment planned
10	No LiPo batteries	Compliant	Only 18650 Li-ion cells to be used
11	Structural: 15g launch, 30g shock survivability	Compliant	FEA analysis in PDR/CDR
12	Descent rate 2–5 m/s (all parts tethered)	Compliant	Parachute sized via simulation
13	Backup COTS altimeter (mandatory)	Compliant	2 altimeters planned (primary + backup)
14	XBEE/LoRa telemetry with NETID = team number	Compliant	NETID set at CDR stage
15	Telemetry command: no transmission until commanded	Compliant	Ground station command protocol
16	Onboard SD card data storage	Compliant	SD module in avionics stack
17	GNSS receiver (NavIC encouraged)	Compliant	NavIC-compatible module to be sourced

18	Ground station with real-time display, CSV export	Compliant	Custom software development planned
19	Battery: 30 min pad wait + flight operation	Compliant	Power budget designed to spec
20	No spring contacts in electrical connections	Compliant	Screw/crimp terminals throughout
21	Team size: 5–8 undergraduate members	Compliant	8 members planned
22	One team per college	Compliant	First and only GITAM team
23	Participation in workshop (mandatory)	Compliant	2 participants registered; ₹25k each
24	Motor retention by positive mechanical means	Compliant	M12 bolt retention system
25	Make in India preference	Compliant	Indigenous components preferred; documented in BOM

7. Declaration & Signatories

We, the undersigned, hereby declare that:

- (a) All information provided in this proposal is accurate and true to the best of our knowledge.
- (b) The design, development, and execution of the rocket system will be carried out entirely by the undergraduate student team, in accordance with IN-SPACE competition rules.
- (c) No external agency will be engaged for design or development (outsourcing is a disqualification offence under IN-SPACE rules).
- (d) All funds granted by GITAM under this proposal will be utilized strictly for competition-related purposes and full financial accountability will be maintained.
- (e) The team commits to representing GITAM with integrity, discipline, and the highest standards of engineering excellence.
- (f) The team is aware that copying designs, codes, or ground station software from any source constitutes a disqualification offence.

Team Leader	Faculty Advisor
Name: _____ Signature: _____ Date: _____	Name: _____ Signature: _____ Date: _____

Head of Department Department of Aerospace Engineering GITAM Deemed to be University	Dean, Research & Innovation GITAM Deemed to be University
Name: _____ Signature: _____ Official Seal: _____ Date: _____	Name: _____ Signature: _____ Official Seal: _____ Date: _____